Flagging in Agrivive

What it is and how it works. Flagging is Agrivive's rule-based trust system (policy *monitoring-v4*). It records verified bad behaviour by buyers and sellers as events, such as a missed pickup, a seller cancelling a confirmed order, a misleading listing, QR misuse, or spam in chat. Each event adds weighted points to the person's score. Only events from the last 90 days count, and excused or overturned events are dropped. The score maps to a level: Good standing, Notice, Warning, Pause 24 hours, Pause 3 days, or Block. A new consequence starts only when a new event moves the person up a level, so re-running the check never repeats a penalty. A single report is only an allegation worth 0 points. It counts after two different reporters give the same reason, or after the system's own records confirm it. No person reviews or decides anything: the system applies the written rules and sends a plain-words notice. Blocked accounts return only through email-code recovery. Sellers get one extra safeguard. A Pause or Block needs both the score and a fault rate of at least 10% over at least 10 orders. Otherwise the seller stays at Warning and is marked "Under watch".

The problem it solves and how it helps. A reservation holds a seller's perishable produce, so a buyer who never shows up, or a seller who cancels or lists something misleading, wastes stock that could have sold [8]. Flagging solves three problems. First, it replaces subjective judgement with consistent, evidence-based rules, since verified transaction events are separated from allegations, which are the fake or manipulated reports that harm peer-to-peer markets [1][2][3]. Second, it makes sanctions proportionate and predictable. Points rise gradually and notices come first, which suits weighted trust evaluation [4] and clear, contestable notices [5][6][7]. Third, it protects honest users. A rating counts only on a completed order, a fault rate stops a few incidents from blocking a high-volume seller, review-bombing and order-flooding are neutralised, and a reasoned excuse or dispute can cancel a penalty. The weights and thresholds are prototype policy settings chosen for evaluation, not values established by the cited studies.

1. Rules, weights and violations

Violation (event kind)	Who	Points	Notes
Inquiry non-response	Seller	1	Reminder at 12 h, warning at 24 h, flag at 48 h
Seller cancels a confirmed order	Seller	2	Also counts in the fault rate
Low rating	Seller	1	1–2 stars on a completed order, and the buyer wrote a reason or it is the 2nd low rating in 30 days
Misleading product info	Seller	2	Confirmed wrong listing
Inappropriate communication	Both	2	
Seller-fault decline at pickup	Seller	2	Seller accepts or stays silent 24 h
QR misuse	Both	3	**Serious: blocks at once**
Missed pickup	Buyer	2	Every time, no free first miss
Late cancellation	Buyer	1	
Invalid decline at pickup	Buyer	1	Reason not allowed
False decline photo	Buyer	2	Photo already used on another decline
Fake reports	Buyer	1	Each group of 3 reports in 14 days with none confirmed
Order from own account	Seller	3	**Serious: refused and blocked at once**
Chat spam	Both	1	Same message 5 times in 10 min pauses sending 10 min. 2nd time in 90 days adds the point.
Personal data in chat	Both	1	Phone or email is hidden. 3rd time in 90 days adds the point.
Review bomb	Reviewers	0	5+ one- or two-star ratings from accounts under 7 days old in 24 h. Ratings ignored, notice sent.
Order flood	Buyer	0	More than 5 orders in 7 days, one buyer to one seller. Extra orders leave seller counts and ranking.

2. Consequences

Score S	Level	Consequence
0–1	Good standing	None
2–3	Notice	Plain-words notice
4–5	Warning	Warning notice
6–7	Pause 24 h	Cannot reserve or transact. Sellers are removed from public listings. Ends automatically.
8–11	Pause 3 days	Same, for 72 hours
12+, or any serious violation	Block	Account set inactive. Restored only by email-code recovery.

Extra seller consequences: at a fault rate of 15% or more (with 10+ orders), the seller is marked "High fault rate" and Priority Boost and Trusted Seller are switched off. Five or more confirmed seller-fault returns raise a *repeat_returns* flag.

3. Formulas

- **Equation 1, trust score:** $S = \Sigma\ (w_k \times n_k)$, over verified, non-overturned events in the last 90 days. w_k is the weight of violation kind k; n_k is its count.
- **Equation 2, seller fault rate:** $F = (C + W + D) \div (O_{done} + O_{cancel} + D + D_{disp}) \times 100$. C = seller cancellations, W = confirmed wrong listings, D = upheld declines at pickup, O_{done} / O_{cancel} = completed / cancelled orders, D_{disp} = disputed declines (denominator only).
- **Level rule:** Block if serious or S ≥ 12; Pause 3d if S ≥ 8; Pause 24h if S ≥ 6; Warning if S ≥ 4; Notice if S ≥ 2; otherwise Good.
- **Seller gate:** a Pause or Block applies only if orders ≥ 10 and F ≥ 10%, or the violation is serious. Otherwise the level drops to Warning.

4. Step-by-step process

1. An event happens: a missed pickup, a cancellation, a low rating, a decline, a chat message, or a report.
2. The system records it as verified or as an allegation. One report is an allegation worth 0 points. Two different reporters with the same reason confirm it.
3. Special rules run first: own-account orders, order flood, review bomb, chat rules, fake-report groups.
4. The score is recalculated: sum the weights of verified events from the last 90 days.
5. For a seller, the fault rate F is calculated and the gate is checked.
6. The raw level is found from the consequences table, then the seller gate may lower it to Warning ("Under watch").
7. If the level is higher than any active action, the system creates one action (notice, warning, pause or block) and sends a notice. If not, nothing repeats.
8. The user may dispute within 7 days, or send an excuse for a missed pickup. The system checks the stored record against the rule. If upheld, the event stops counting and the level is recalculated.
9. Pauses expire on their own. Events older than 90 days fade out of the score.

5. Situation example with computation

Seller Mang Ben, last 90 days:

Event	Count × weight	Points
Inquiry non-response	1 × 1	1
Seller cancellation	1 × 2	2
Low rating with reason	1 × 1	1

Event	Count × weight	Points
Misleading product info	1 × 2	2
Seller-fault decline	1 × 2	2
S		**8**

Raw level: S = 8 gives **Pause 3 days**.

Fault rate: C = 1, W = 1, D = 1, so faults = 3. Orders are 25 completed + 1 cancelled + 1 upheld decline + 0 disputed = 27. F = 3 ÷ 27 × 100 = **11.1%**.

The gate is met (27 ≥ 10 orders and 11.1% ≥ 10%), so Mang Ben **is paused for 72 hours** and removed from public listings. F is below 15%, so there is no "High fault rate" flag.

Same score, busier seller. With 60 total orders, F = 3 ÷ 60 = 5%. The gate fails, so the level drops to **Warning** and Mang Ben is "Under watch", with no pause.

Buyer Ana. A missed pickup (+2) and a late cancellation (+1) give S = 3, so Ana gets a Notice. Another missed pickup (+2) gives S = 5, a Warning. A third (+2) gives S = 7, a **24-hour pause** during which Ana cannot reserve. An accepted excuse (within 24 h, once per 30 days) would cancel one of those missed pickups and recalculate her level.

The numbers in this example are a sample, not real data.

Conclusion

Flagging gives Agrivive one transparent, system-decided way to keep reservations dependable. Verified behaviour becomes weighted points over a rolling 90-day window. Points map to graduated consequences, from notices to pauses to blocks. Seller fault rates, 7-day disputes, excuses and anti-abuse rules (order flood, review bomb, fake reports, chat spam) protect honest users from unfair penalties. Allegations stay at zero points until the evidence confirms them. The numbers are tunable prototype settings that the owner still needs to test and tune. Even so, the structure follows research on reputation systems, review integrity and fair notice, and it keeps produce reservations trustworthy for both buyers and sellers.

References

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Only [5], [6] and [7] were checked against their arXiv pages. The other entries come from search results or the capstone bibliography: confirm authors, volumes and pages before submitting.